Mst Afsana Mimi

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Professional Summary

Data analyst and Statistics master's graduate experienced in cleaning, validating, analyzing, and organizing large datasets using SQL, Python, and R. Skilled in descriptive analysis, trend identification, reproducible reporting workflows, and communicating statistical findings clearly. Brings hands-on experience with Tableau, data-quality checks, and collaborative analytical projects.

Technical Skills

Analysis and reporting: Data cleaning, data validation, descriptive analytics, trend and anomaly identification, statistical analysis, data visualization
Programming and querying: SQL (joins, aggregations, window functions), Python (Pandas, NumPy, Scikit-learn), R
Tools and platforms: Tableau, Git/GitHub, AWS S3 and EC2 (familiarity), ETL/ELT workflows
Methods: Regression, hypothesis testing, statistical modeling, model evaluation

Experience

Adjunct Faculty, Department of Psychological and Brain Sciences Sep 2026–Present
University of Massachusetts Amherst *Amherst, MA*

Teaching Assistant, Data and Statistics Sep 2023–May 2026
University of Massachusetts Amherst *Amherst, MA*

- Supported multiple student projects involving data preparation, analysis, and interpretation using R and Python.
- Reviewed analytical work for accuracy and helped identify errors in data processing and statistical logic.
- Explained statistical findings and technical concepts clearly to audiences with different levels of analytical experience.

Research Assistant, NLP and Data Analysis Jul 2021–Jun 2022
University of Dhaka *Dhaka, Bangladesh*

- Processed and organized large-scale text datasets through reproducible data-ingestion and transformation pipelines.
- Cleaned and validated analytical datasets, performing consistency checks and investigating data-quality issues.
- Built preprocessing, feature-engineering, and evaluation workflows in Python and summarized analytical results for research collaborators.

Research Assistant, Socioeconomic Data Analysis Jul 2018–Jun 2019
University of Dhaka *Dhaka, Bangladesh*

- Cleaned, transformed, and structured large survey datasets into analysis-ready formats.
- Applied descriptive and statistical analyses to identify socioeconomic trends and support policy research.
- Maintained reproducible code and analysis workflows with Git to support collaboration and consistent outputs.

Selected Projects

Relational Database Analytics Framework *Python, relational data*

- Develop a framework that interprets database schemas, constructs relational evidence, and supports predictive analysis of structured data.
- Evaluate workflows on realistic database tasks and examine the accuracy and consistency of analytical outputs.

High-Dimensional Feature Selection *Statistical analysis*

- Designed and evaluated a feature-selection method for high-dimensional censored data using simulations and biomedical datasets.

Education

University of Massachusetts Amherst — MS, Statistics; GPA: 3.77 May 2026
University of Dhaka — MS, Applied Statistics; GPA: 3.75 Dec 2018
University of Dhaka — BS, Applied Statistics; GPA: 3.61 Dec 2016

Publication and Recognition

Publication: Mimi, A., & Khan, M. H. R. (2021). "Variable selection for censored data using modified correlation adjusted correlation (MCAR) scores." *Statistics in Medicine*, 40(23), 5046–5064.
Recognition: ISCB Conference Award for Scientists; National Science and Technology Fellowship; Dhaka University Merit Scholarship.